



NSF FUTURE MANUFACTURING DATA CHALLENGE

Predicting Local Laser-Track Geometry from Thermal Video

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4 LASER POWERS

6 GEOMETRY DESCRIPTORS

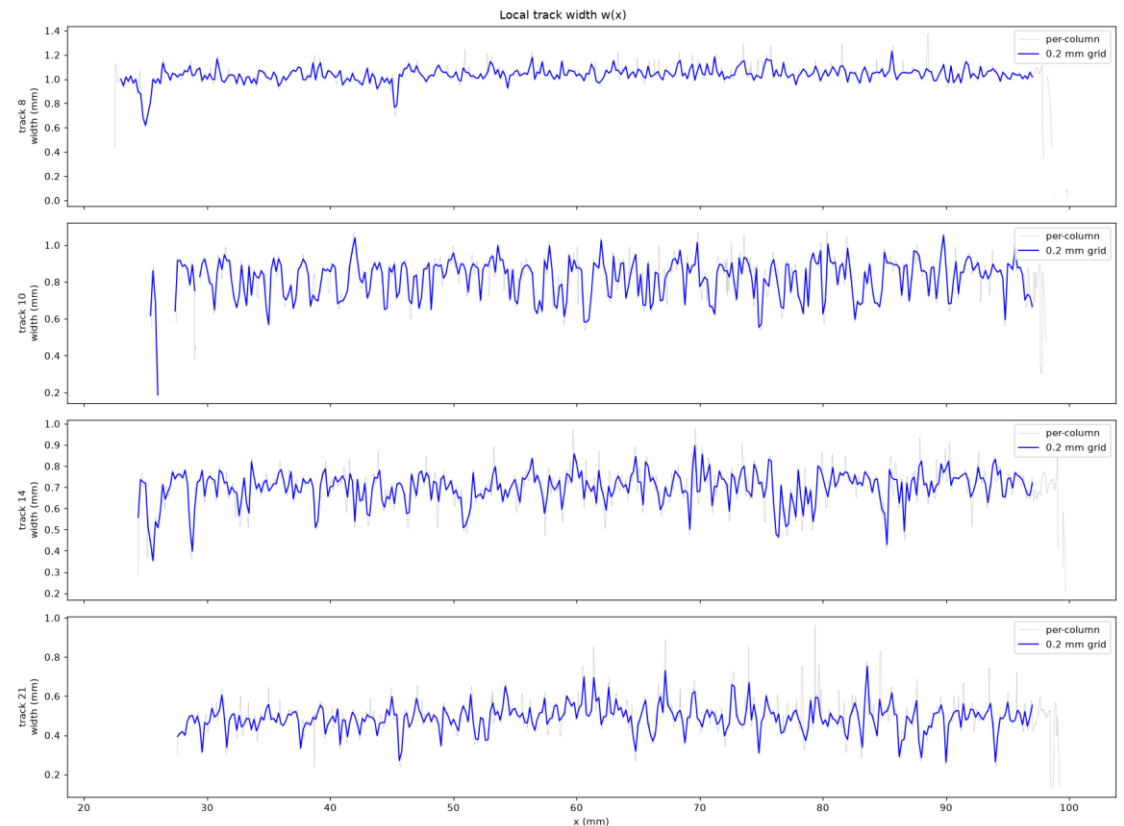
CALIBRATED
UNCERTAINTY

THE PROBLEM

One average width hides the defects that matter most

Even when laser power, speed, and every process setting stay fixed, a laser track is not uniform. It gets wider here, narrower there, wavier somewhere else along its length.

A single track-average number hides exactly the local weak points that cause real manufacturing failures. This project predicts geometry as a function of position along the track — not one number per track.



Measured local width along all 4 tracks — real data, not illustrative

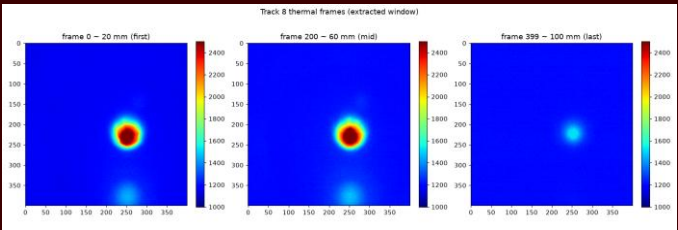
INPUTS

Three aligned measurements, per physical location

1

Thermal Video

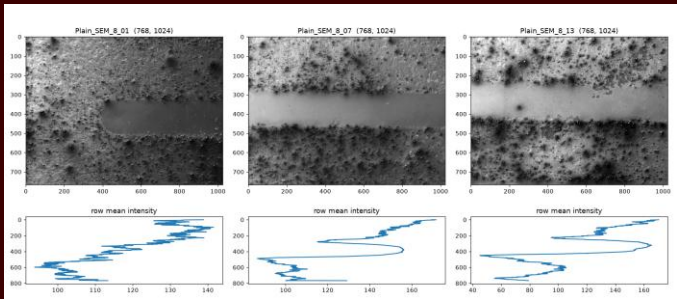
50 fps melt-pool camera, 400×400 px, ~14 μm/px



2

SEM Substrate

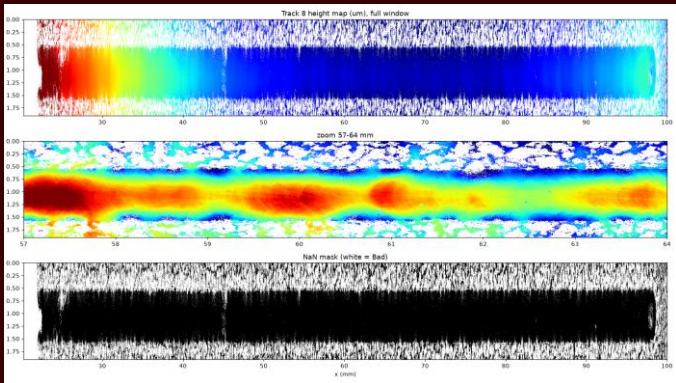
Close-up surface imagery around the track (masked)



3

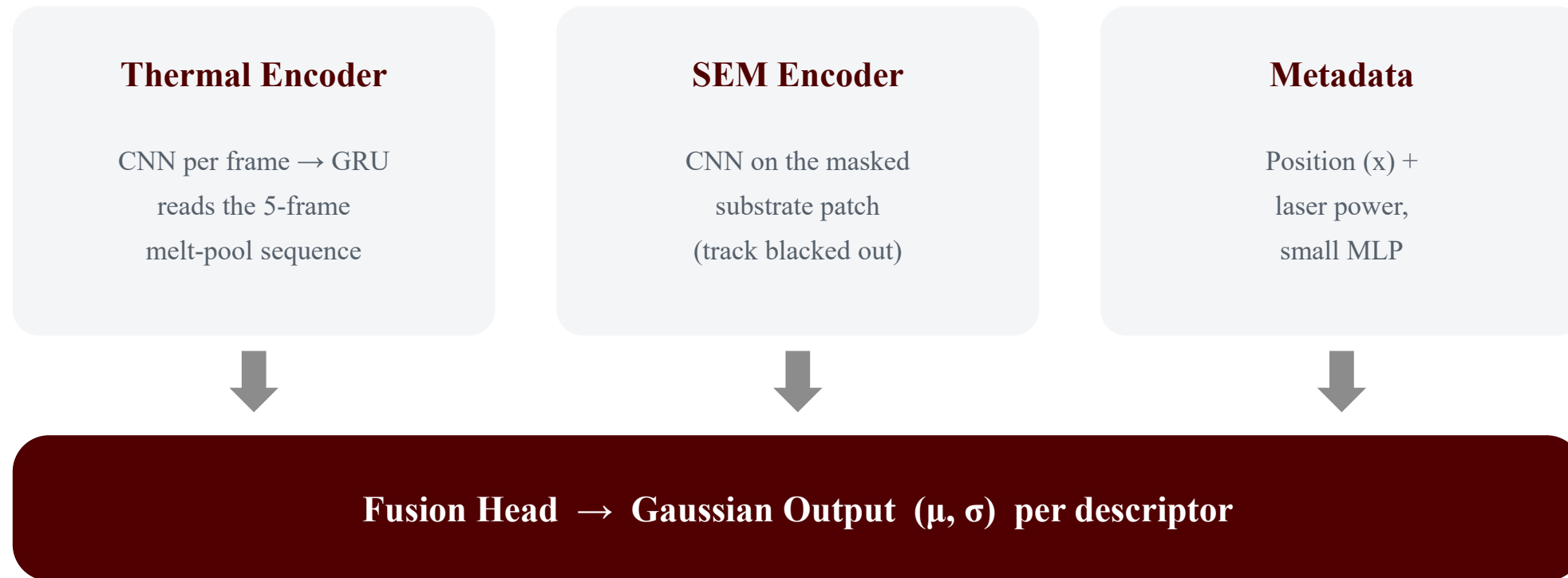
Height-Map Truth

Wyko profilometry — the ground truth being predicted



MODEL

A compact, three-branch probabilistic network



~200,000 parameters — small enough to train on CPU, sized for a 4-track dataset. out_dim = 1 for a single descriptor, or 6 for the joint model predicting width, both boundaries, contour deviation, waviness, and roughness at once.

EVALUATION

Leave-one-track-out, across all four laser powers

Each track takes a turn as the fully unseen test case. The model trains on the other three tracks and is scored on a laser power it has never encountered.



Track 21 (200 W) is the hardest fold: it extrapolates below every training power and has the noisiest ground truth — exactly the organizers' suggested robustness test.

RESULT

Primary width prediction

170

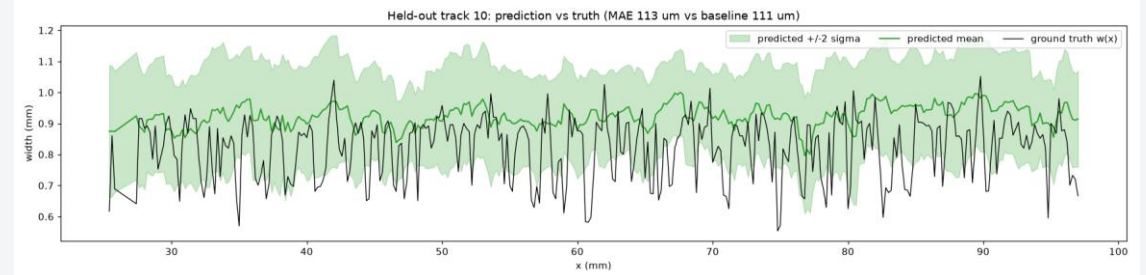
μm mean error

27%

better than constant baseline

Compared against the fair baseline: guessing the training-set average width every time, with no image data at all. All four laser powers rotate through the held-out position; results below are the calibrated, leave-one-track-out mean.

The gain is uneven, and concentrated at the extreme powers: +35% at 400 W and +44% at 200 W, but only +1% at 350 W and -40% at 300 W.



Predicted width (green) vs. measured truth (black), held-out Track 10

RESULT

Six geometry representations, compared head-to-head

Each descriptor modeled two ways — a dedicated specialist, and one joint model predicting all six at once — to find the most accurate method for each.

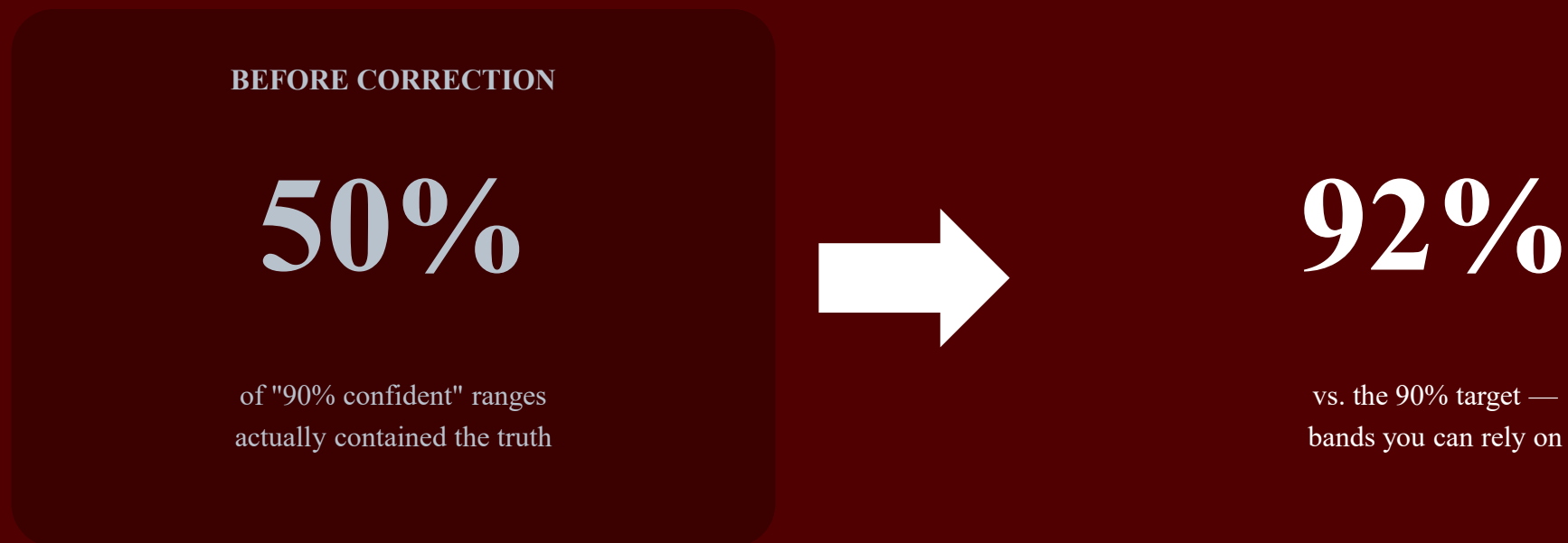
Descriptor	Best method	MAE	vs. baseline
Width	Scalar	170 μm	+27.0%
Left boundary	Joint (vector)	96 μm	+15.2%
Right boundary	Scalar	112 μm	+27.6%
Contour deviation	Joint (vector)	30 μm	−12.6%
Waviness	Scalar	16.9 μm	−16.3%
Roughness	Joint (vector)	15.3 μm	−15.0%

Width and both boundaries clearly beat baseline; the three fine-scale descriptors remain an open challenge on unseen laser powers, despite excellent calibration.

UNCERTAINTY

From overconfident to trustworthy

Every prediction reports a range, not just a guess. A leak-free correction — fit only on OTHER folds, never the track being tested — fixes systematic overconfidence.

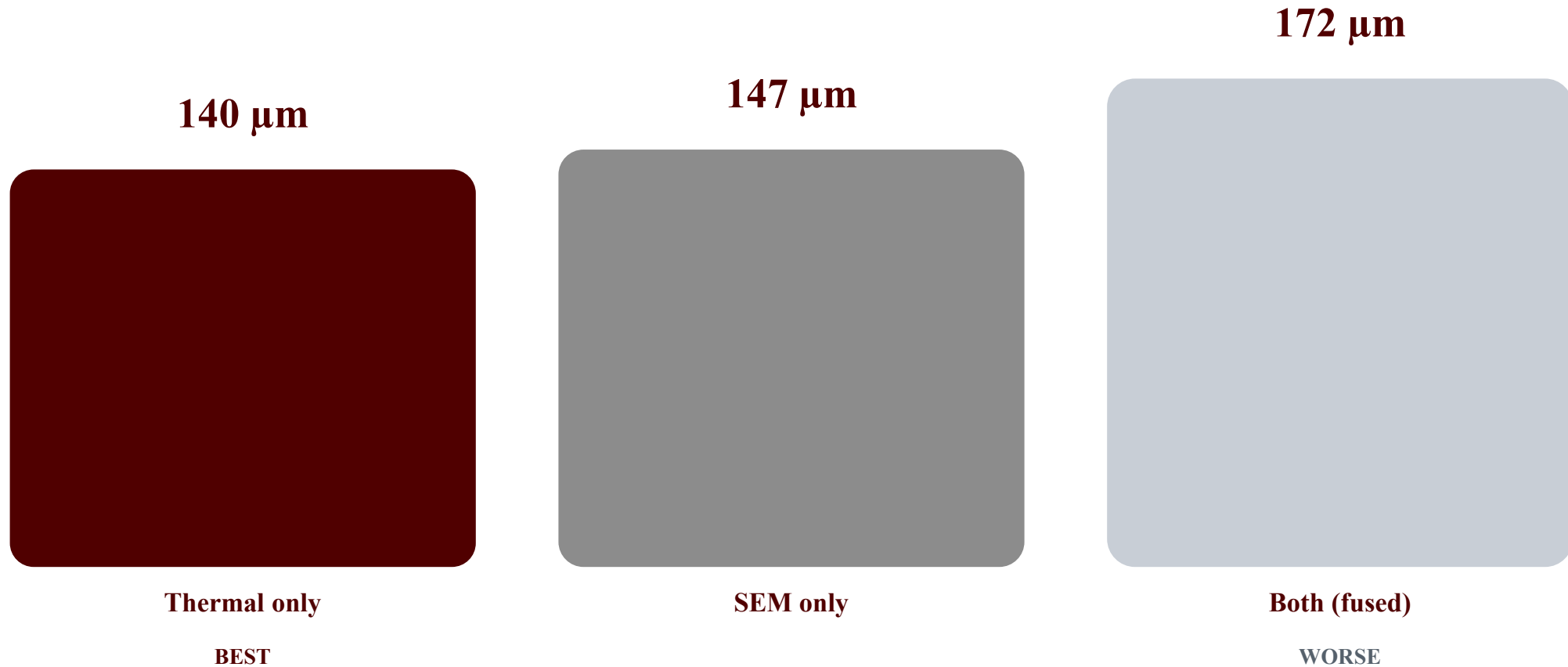


Additive bias-floor correction: widens uncertainty by a floor, the right shape for systematic (not random) error. The mean prediction itself is unchanged.

COUNTERINTUITIVE FINDING

More sensors made the prediction worse

The width model trained three ways — thermal-only, SEM-only, and both fused — identical folds, seed, and epochs otherwise.



Both single-modality models individually beat the fused model. With only four tracks of data, more inputs meant more ways to overfit, not more signal.

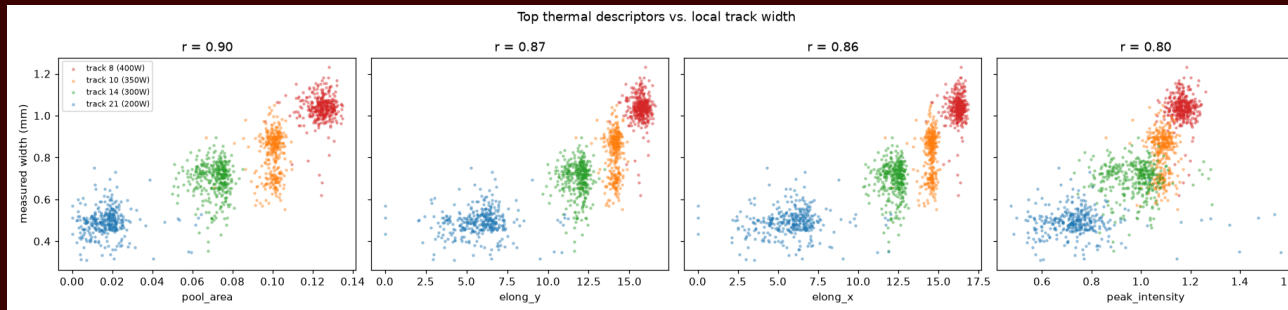
Which melt-pool features actually matter

$$r = 0.90$$

pool size correlates with width across powers — but drops to ~0 within a single track. Size explains the big picture, not local wiggle.

$$r = 0.71$$

melt-pool vertical asymmetry (centroid_dy) is the strongest predictor of where the MODEL gets it wrong — a live, physically grounded warning sign.



Melt-pool descriptors vs. measured width, colored by laser power

CONCLUSION

Four takeaways

1

Between powers, the link is strong

Melt-pool size predicts width ($r=0.90$); the model beats baseline by 27% on average across unseen laser powers.

2

Within a track, we don't yet resolve it

Detrended prediction-vs-truth correlation is ~ 0 (mean $r=-0.03$) — the model captures the power-driven level, not local detail.

3

Process and substrate both matter

Both carry real, largely independent signal; substrate dominates specifically at the lowest power.

4

Asymmetry is an early-warning signal

Melt-pool asymmetry ($r=0.71$ with model error) tells you when NOT to trust a prediction — insight beyond raw accuracy.

Thank you

Full technical report, plain-language report, executable notebook, and complete source code are available in the project repository.

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